Engine Model(s): PT6A

70-00-00

STANDARD PRACTICES - REPAIR

1. General

A. The primary purpose of repair is to enable a component or engine part to be reworked to a condition which will provide safe operation. Approved repair schemes are detailed in the relevant sections of this manual using the various consumables listed in CONSUMABLE MATERIALS section.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC03-001	Oil, Engine Lubricating
PWC05-061	Cloth, Abrasive
PWC05-147	Electrode, Welding
PWC05-161	Solution, Touch-up
PWC05-197	Acid, Chromic
PWC11-014	Alcohol, Isopropyl

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name	Application
PWC30271-100	Spreader, Nut	Obsolete - replaced by PWC31771
PWC31771	Spreader, Nut	Alternate to PWC30271-100

4. Fixtures, Equipment and Supplier Tools

The fixtures, equipment and supplier tools listed below are used in the following procedures.

Name	Remarks
Welder	AC/DC with high frequency controls
Torch	Linde HW-10 or equivalent with recirculating water
Regulator	Argon gas
Flowmeters	(2 off)

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5. Welding

CAUTION: WELDING MUST BE DONE BY QUALIFIED PERSONNEL ONLY, USING SAFE WELDING PRACTICES, ADEQUATE VENTILATION, AND PROTECTIVE EQUIPMENT. DO NOT WELD IN CONFINED AREAS WITHOUT PROPER VENTILATION.

A. General

The following procedures apply to components that have been removed from the engine. Weld repair of components still installed on the engine, is not permitted under any circumstance.

A tungsten inert gas fusion welding process must be used when repairing cracks in engine parts. Argon or helium gas may be used in inert gas welding; however, argon is preferred by Pratt & Whitney Canada because its greater density reduces its rate of diffusion with the atmosphere.

Inert gas welding is a gas arc welding process which uses inert gas to protect the weld area from the surrounding atmosphere. The heat necessary for welding is provided by a very intense electric arc which is struck between a non-consumable thoriated tungsten (tungsten and thorium alloy) electrode and the metal part. On repairs where a filler material is required, a welding rod, of appropriate material, is fed into the weld area and melted with the base metal in the same manner as in conventional gas welding.

B. Cleaning Before Welding

(1) Refer to CLEANING.

C. Welding Procedures

WARNING: IF AN OXYGEN REGULATOR IS USED WITH ARGON GAS, DO NOT REINSTALL REGULATOR ON AN OXYGEN CYLINDER. ARGON GAS CONTAINS OIL AND THE POSSIBILITY EXISTS OF AN EXPLOSION DUE TO THE PRESENCE OF OIL IN THE REGULATOR. USE SAFE WELDING PRACTICES, ADEQUATE VENTILATION AND PROTECTIVE EQUIPMENT. DO NOT WELD IN CONFINED AREAS WITHOUT EXHAUST VENTILATION.

NOTE: The following procedures apply to components that are dispositioned for repair by welding and are removed from the engine. Weld repair of components still installed on an engine is not permitted.

(1) Welding must be done only by certified personnel.

(a) Connect one argon gas flowmeter to torch and connect other flowmeter, if necessary, behind crack as a backup when ready to weld.

(b) Connect torch to arc welder for straight polarity.

(c) Insert a $^{1}/_{6}$ inch diameter thoriated tungsten electrode (PWC05-147) into torch, allowing approximately $^{5}/_{16}$ inch of electrode to protrude from ceramic cup. The ceramic cup used must not be smaller than a No. 6 for the HW-10 torch, in order to provide a sufficient amount of inert gas to the weld area where cleanliness of the surface beader is important.

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NOTE: A plain tungsten electrode may be used, but a needle point must be maintained on the electrode for more positive starting of arc. Where possible, the electrode must be kept clean of material pickup.

(d) If it is necessary to protect underside of weld from contamination, possible weld porosity, or poor surface finish, attach argon gas line behind crack to be welded. This may be accomplished by fabrication of small box-like fixtures having edges shaped to fit contours of part to be welded and using a suitable connection to secure gas line. Seal fixture to part in order to conserve gas leakage to minimum.

NOTE: The amount of gas pressure to provide sufficient backup must be determined. This pressure should not allow the gas to stir the weld puddle. Do not use flux when welding in an inert atmosphere.

(e) When high frequency controls are used, it is not necessary to strike an arc. Hold torch vertically to work so that the shielding gas (argon) forms a protective envelope around the weld. It is preferable to have area to be welded in a horizontal position.

(f) When high frequency controls are not used, strike arc and hold electrode tip approximately $^{1}/_{8}$ to $^{1}/_{4}$ inch above surface to be welded until a puddle is formed. Add welding rod, as applicable, and proceed as in conventional gas welding.

NOTE: Test welds, using corresponding material of same thickness and joint design, should be made to determine correct gas pressure and arc welding settings.

(g) Because a high percentage of steel parts used in the engine are fabricated from 12 percent chromium corrosion-resistant steels, which are characterized by their susceptibility to air hardening, field repair of cracks by fusion welding is a special problem. The high temperature at which fusion weld repairs are made and the subsequent air cooling of the part, or parts, from these temperatures usually results in an increase in material hardness and a loss in ductility. Parts on which fusion weld repairs have been made have a tendency to crack because the steel structure becomes unstable, brittle and highly stressed. The structure of the material can be improved by reheating the parts and controlling the cooling rate.

D. Cleaning After Welding

(1) Refer to CLEANING.

E. Local Stress Relief

(1) Components which are not highly stressed may be repaired by fusion welding. It is possible to partially restore the original properties of such welded parts through use of local heat treatment. The localized heat may be applied by neutral flame of an oxyacetylene torch. Extend the stress relief one inch minimum beyond the welded area. After the desired heat has been applied to the part for the correct length of time, reduce temperature of part gradually.

F. Inspection of Welding

(1) Inspect repair area (Ref. INSPECTION).

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6. Blend Repairs

A. Repair Procedure

(1) Blend repair using carborundum stone.

CAUTION: DO NOT POLISH WITH POWER TOOLS.

(2) Polish blended area using crocus cloth (PWC05-061).

(3) Blend all repairs and finish smoothly.

(4) Lines, scratches or sharp edges which may cause a concentration of stress are not permitted.

B. Definitions

BLENDING	An operation that removes an irregularity or imperfection from a surface and results in restoring the surface to a smooth acceptable condition.
GRINDING	An operation that removes material by the use of an abrasive material to produce a predetermined size.
POLISHING	A finishing operation which produces a smooth surface finish.

C. General

Repairs are defined as freehand blending, deburring and/or polishing. It does not include machining or machine grinding.

Repair only defined damage within the limits specified in the relevant Chapter/Section/Unit. Undefined damage shall render the component unserviceable.

Do repairs in a direction so that the finishing marks follow the lay of the original manufacturing machining marks.

Restore surface finish of repaired area to the original manufacturing finish.

Make sure repaired area conforms to the original manufacturing shapes, profiles, contours and radii.

Make sure local repairs extend over an area 10 times the depth of damage.

Maintain a smooth transition between repaired areas and adjacent areas.

Make sure repaired area is 1/3 deeper than the damage to ensure removal of damage.

Examine all repaired components.

D. Air Gun

The air gun is hand held, with an adjustable spindle speed and drives a suitable mandrel.

Use a piece of abrasive cloth, rotary file, abrasive disk or wheel of appropriate shape, dimensions and grit size for hand grinding rotating components.

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Adjust the rotation of the hand blending wheel to obtain the required surface finish.

Do not allow the wheel to exceed 2000 RPM to prevent local overheating of the material.

Adjust the applied pressure of the wheel to produce a maximum of 0.0005 inch of material removal in each pass.

A light, equal pressure on the wheel gives the best results as it permits the abrasive to cut easily without loading up and overheating.

7. Threaded Inserts

A. Replacement of Threaded Inserts

CAUTION: BEFORE USING INSERTS, IN ANY PART OF ENGINE, REFER TO ILLUSTRATED PARTS CATALOG FOR APPROVED LOCATIONS AND PART NUMBER OF INSERT.

(1) Remove unserviceable insert using an approved extraction tool.

(2) Clean out hole and make sure swarf and other foreign matter is removed.

(3) Install new threaded insert into threaded hole using installation tool. Insert outer thread must be between one and one-half threads below surface of hole or counterbore, whichever applies.

(4) Cut off driving tang at notch using approved tang removal tool and remove tang from holes.

(5) Inspect repaired hole.

8. Studs

A. Replacement of Studs

CAUTION: BEFORE INSTALLING NEW STUDS, REFER TO ILLUSTRATED PARTS CATALOG FOR APPROVED LOCATIONS AND PART NUMBERS.

When necessary, oversize studs are to be installed in place of those which are broken, stretched, loose or have damaged external threads. Whenever a stud which is already oversize requires replacement, install the next oversize stud. Use correct stud drivers to install new studs and torque to appropriate value (Ref. Tables 801 or 802, as applicable).

Before installing new studs, refer to associated Illustrated Parts Catalog, for approved locations and part numbers. When threads of a stud hole are damaged beyond dimensions suitable for fitting a maximum oversize stud, it is usually possible to do repairs by the installation of a helical coil (Ref. Subpara. A.) or key-type insert. Consideration should, however, be given to reduction of thickness of walls of parent metal around insert and the required strength. Specific problems should be referred to P&WC Service Department (Ref. INTRODUCTION).

B. Replacement of Studs

(1) Remove damaged stud using an approved method.

CAUTION: BEFORE INSTALLING NEW STUDS, REFER TO ILLUSTRATED PARTS CATALOG FOR APPROVED LOCATIONS AND PART NUMBERS.

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(2) Examine stud hole for condition; on worn stud holes use oversize studs.

(3) Install stud using an approved stud driver to correct protrusion height; make sure torque limits (Ref. Tables 801 or 802, as applicable) are not exceeded.

Table 801 Standard Stud Torque Limits

Stud Thread Size (Drive End)	Minimum	Maximum Necked Down $	Maximum Plain #
0.112-40	4	-	8
0.138-32	8	-	14
0.164-32	10	30	30
0.190-24	15	40	45
0.216-24	20	65	70
0.250-20	40	95	105
0.3125-18	85	210	230
0.375-16	160	375	425
0.4375-14	200	600	675
0.500-13	250	950	1050
NOTE: 1. Symbol $: These limits apply where the unthreaded diameter of the stud is less than the minimum minor diameter of the coarse pitch thread (drive end).			
NOTE: 2. Symbol #: These limits apply where the unthreaded diameter of the stud is equal to or greater than the minimum minor diameter of the coarse pitch thread (drive end).			

Table 802 Stepped Stud Torque Limits

Stud Thread Size (Nut End)	Minimum	Maximum Necked Down $	Maximum Plain #
0.138-40	8	-	18
0.164-36	10	30	30
0.190-32	15	45	50
0.216-28	20	65	75
0.250-28	40	115	125
0.3125-24	85	240	260

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Stud Thread Size (Nut End)	Minimum	Maximum Necked Down $	Maximum Plain #
0.375-24	160	450	500
0.4375-20	200	700	800
0.500-20	250	1150	1300
NOTE: 1. Symbol $: These limits apply where the unthreaded diameter of the stud is less than the minimum minor diameter of the fine pitch thread (nut end).			
NOTE: 2. Symbol #: These limits apply where the unthreaded diameter of the stud is equal to or greater than the minimum minor diameter of the fine pitch thread (nut end).			

C. Repair of Damaged Stud Hole

(1) Make sure damaged hole is suitable for repair (Ref. Subpara. B. preceding).

(2) Measure core depth of existing hole.

(3) Select relevant size drill and drill hole to measured depth (Ref. Step (2), preceding).

(4) Using relevant special tap, thread hole one thread deeper than insert to be fitted.

(5) Where applicable, counterbore hole to required angle and depth.

(6) Clean out hole and ensure freedom from metal chippings and other foreign matter.

(7) If part being repaired is magnesium, treat tapped hole with chrome pickle touch-up solution (PWC05-161):

(a) Clean area to be treated with abrasive cloth (PWC05-061) and/or cloth impregnated with isopropyl alcohol (PWC11-014).

(b) On rough surfaces, remove corrosion using wire brush or abrasive paper.

(c) On finished surfaces, remove corrosion using swab or brush impregnated with solution (PWC05-197) at 85° to 93°C (180° to 200°F).

(d) Rinse with clean, cold water.

(e) Rinse with clean, hot water.

(f) Dry using clean, dry compressed air.

(g) Apply solution (PWC05-197) to area to be treated using a clean cloth. Repeat swabbing at frequent intervals to make sure treated area is maintained wet with solution during treatment.

(h) Thoroughly rinse touched up area by swabbing several times with clean cloth moistened with water.

(i) Dry using clean, dry compressed air.

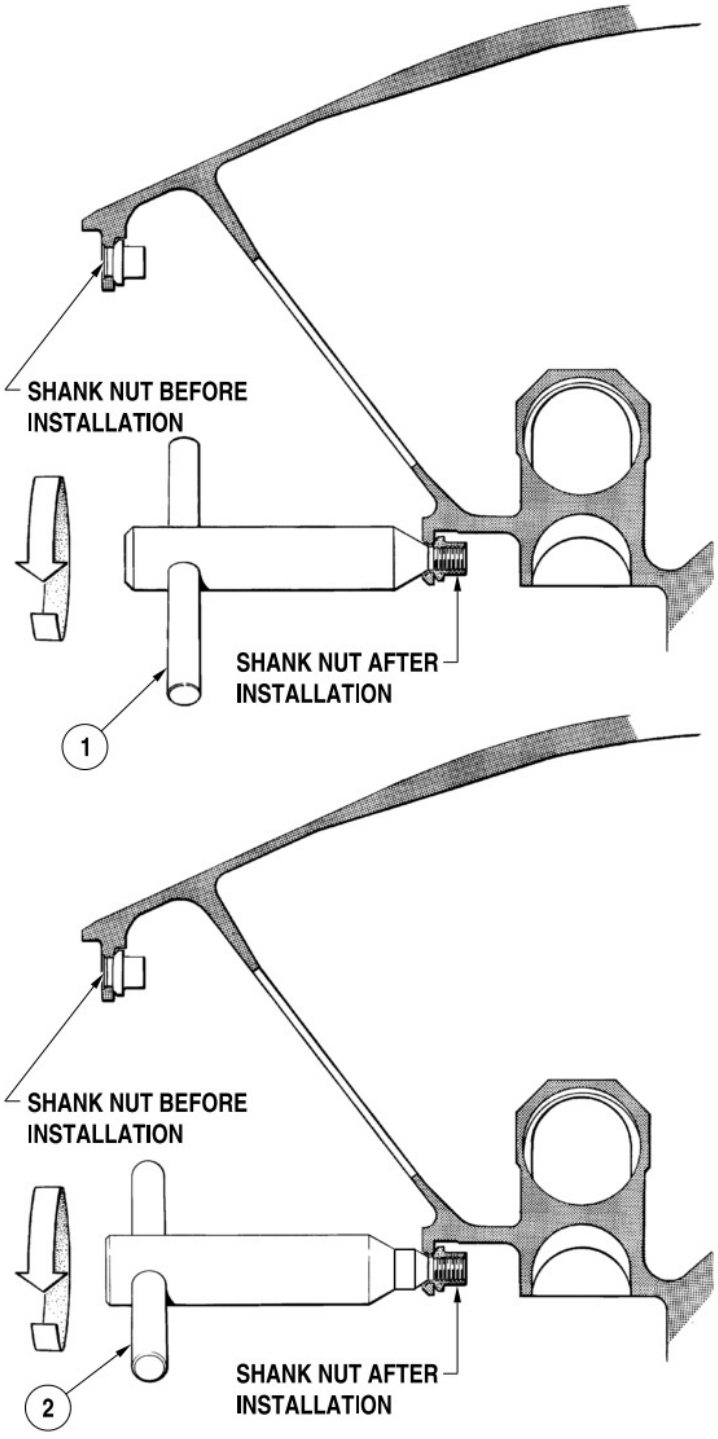
(8) Using appropriate size threaded insert, install insert into repaired stud hole (Ref. Para. B.).

9. Shanknuts

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A. Replacement of Shanknuts (Ref. Fig. 801)

Figure 801 Shanknut Replacement



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1. Spreader (PWC30271-100)
2. Spreader (PWC31771)

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CAUTION: AVOID DAMAGE TO FLANGE.

(1) Using a suitable drill, partially remove the flared end of shanknut.

(2) Remove shanknut body using a parallel pin punch at drilled end to shear weakened flare, and retrieve the shanknut case.

(3) Install new shanknut (Ref. IPC) and hold against flange.

(4) Lightly lubricate tapered portion of spreader (1) (PWC30271-100) or (2) (PWC31771) with oil (PWC03-001) and screw into shanknut until shank end is flared against flange.

(5) Remove spreader and examine flared end of shanknut for correct forming with no evidence of deformation or cracks.

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